Project Design Phase
Proposed Solution Template

Date	03 April 2026
Team ID	SWTID-2026-8629
Project Name	Smart Library
Maximum Marks	2 Marks

Model Performance Testing – Smart Library System

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1. Model Summary

The system is developed as a web-based application for managing library operations efficiently. It includes modules such as user authentication, book management, issue and return tracking, and report generation.
The backend handles business logic and database operations, while the frontend provides an interactive interface for students and librarians. The system ensures secure data handling and efficient performance.

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2. Accuracy

Since this project is a system-based application (not a machine learning model), accuracy is interpreted in terms of system performance and correctness.

- **Training Accuracy:** Not Applicable (No ML model used)
- **Validation Accuracy:** System tested with multiple inputs and showed correct outputs with high reliability (approx. 95–100% functional accuracy)

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3. Fine Tuning Result (if Done)

System performance was improved by:

- Optimizing database queries
- Improving UI responsiveness

- Fixing bugs identified during testing
- **Validation Accuracy:** Improved system efficiency and reduced response time after optimization (near 100% correct functionality)

Conclusion

The Smart Library System performs efficiently with accurate data processing and reliable output. Testing confirms that all modules (book management, issue/return, and reporting) function correctly, and the system meets the required performance standards.